Unit: 6 Unit 6: Systems of Linear Equations and Inequalities **Name:** ____________
Topic: Applications of Systems of Inequalities **Date:** __________

Homework (Habanero)

Q1. A hospital wants to purchase x units of Medicine A and y units of Medicine B. The cost of Medicine A is \$5 per unit and the cost of Medicine B is \$3 per unit. The total budget is \$150. The inequality that models this situation is:

(A) $5x + 3y \leq 150$
(B) $5x - 3y \geq 150$
(C) $5x + 3y \geq 150$
(D) $3x + 5y \leq 150$
(E) $3x - 5y \leq 150$

Q2. In a biology experiment, the growth rate of cells is modeled by the inequality $2x + 5y \geq 12$, where x is the number of hours and y is the number of cells. What does this inequality represent?

(A) The minimum number of cells required after a certain number of hours
(B) The maximum number of cells that can be grown in a certain number of hours
(C) The minimum number of hours required to grow a certain number of cells
(D) The maximum number of hours required to grow a certain number of cells
(E) The average number of cells grown per hour

Q3. A pharmaceutical company produces two types of medication: x units of Medication X and y units of Medication Y. The production cost of Medication X is \$2 per unit and the production cost of Medication Y is \$4 per unit. The total production cost is \$240. The inequality that models this situation is:

(A) $2x + 4y \leq 240$
(B) $2x - 4y \geq 240$
(C) $2x + 4y \geq 240$
(D) $4x + 2y \leq 240$
(E) $4x - 2y \leq 240$

Q4. A health clinic has x doctors and y nurses. The clinic wants to ensure that the number of doctors is at least half the number of nurses. The inequality that models this situation is:

(A) $x \geq \frac{1}{2}y$
(B) $x \leq \frac{1}{2}y$
(C) $x \geq 2y$
(D) $x \leq 2y$
(E) $x = \frac{1}{2}y$

Q5. A researcher is studying the effect of two variables, x and y, on the growth of bacteria. The researcher finds that the growth rate is modeled by the inequality $x+2y \leq 10$. What does this inequality represent?

(A) The maximum growth rate of bacteria
(B) The minimum growth rate of bacteria
(C) The average growth rate of bacteria
(D) The maximum number of bacteria that can grow
(E) The minimum number of bacteria required for growth

Q6. A hospital wants to purchase x units of medical equipment and y units of supplies. The cost of medical equipment is \$10 per unit and the cost of supplies is \$5 per unit. The total budget is \$500. The inequality that models this situation is:

(A) $10x + 5y \leq 500$
(B) $10x - 5y \geq 500$
(C) $10x + 5y \geq 500$
(D) $5x + 10y \leq 500$
(E) $5x - 10y \leq 500$

Q7. In a medical study, the dosage of a new medication is modeled by the inequality $3x + 2y \geq 12$, where x is the number of milligrams and y is the number of doses. What does this inequality represent?

(A) The minimum dosage required for the medication to be effective
(B) The maximum dosage allowed for the medication
(C) The average dosage required for the medication
(D) The minimum number of doses required for the medication
(E) The maximum number of milligrams allowed per dose

Q8. A pharmaceutical company produces x units of Medication A and y units of Medication B. The profit from Medication A is \$3 per unit and the profit from Medication B is \$2 per unit. The total profit is \$150. The inequality that models this situation is:

(A) $3x + 2y \geq 150$
(B) $3x - 2y \leq 150$
(C) $3x + 2y \leq 150$
(D) $2x + 3y \geq 150$
(E) $2x - 3y \leq 150$

Open Ended Questions (FRQ)

Q9. A hospital wants to purchase medical equipment and supplies for a new wing. The cost of medical equipment is \$8 per unit and the cost of supplies is \$4 per unit. The total budget is \$320. If the hospital wants to purchase at least 10 units of medical equipment and at least 20 units of supplies, write an inequality to model this situation and find the feasible region.

Q10. A researcher is studying the effect of two variables, x and y, on the growth of a certain type of cell. The researcher finds that the growth rate is modeled by the inequality $2x + 3y \leq 15$. If the researcher wants to find the maximum growth rate, write an equation to model this situation and find the maximum value of x and y.

Chef's Tips

- Tip 1: Ensure students understand the context of the problem before solving the inequality.
- Tip 2: Encourage students to graph the inequality to visualize the feasible region.
- Tip 3: Remind students to consider the constraints of the problem when finding the feasible region.